

Advanced Statistics & probability

Lecture 2

By

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Chapter (2)

Student distribution (T-distribution)

❑ The T-distribution (or Student's t-distribution) is a continuous probability distribution similar to the normal distribution but with heavier tails, used for estimating population parameters when:

- The sample size is small (typically $n < 30$).
- The population standard deviation is unknown.
- The data are approximately normally distributed.

❑ The T- distribution shares some characteristics of the normal distribution and differs from it in others. The T- distribution is similar to the standard normal distribution in these ways:

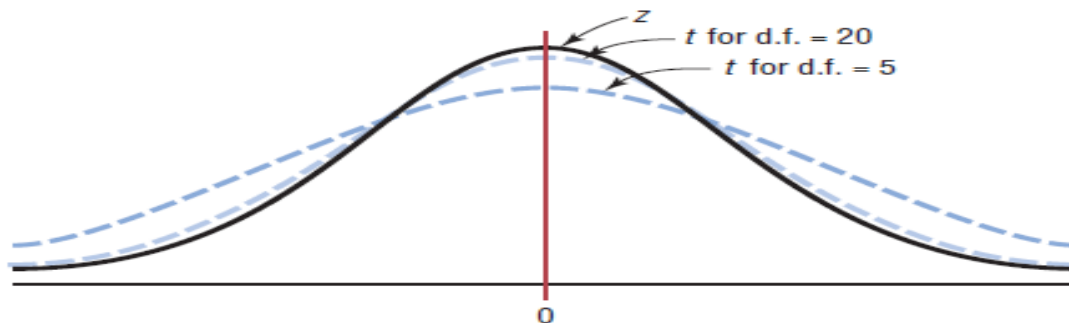
- 1) The support of T- distribution is from $-\infty$ to ∞ .
- 2) It is bell-shaped.
- 3) It is symmetric about the mean.
- 4) The mean, median, and mode are equal to 0 and are located at the center of the distribution.
- 5) The curve never touches the x axis.
- 6) T- distribution distributions are denser in the center and less dense in the tails.

❑ The T- distribution differs from the standard normal distribution in the following ways:

- 1) The variance is greater than 1.
- 2) The t distribution is actually a family of curves based on the concept of degrees of freedom, which is related to sample size.
- 3) As the sample size increases, the t distribution approaches the standard normal distribution.

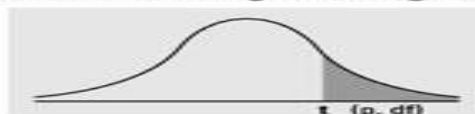
❑ Remark:

The degrees of freedom are the number of values that are free to vary after a sample statistic has been computed.



T-Distribution Table:

Numbers in each row of the table are values on a t -distribution with (df) degrees of freedom for selected right-tail (greater-than) probabilities (p).



df/p	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.0005
1	0.324920	1.000000	3.077684	6.313752	12.70620	31.82052	63.65674	636.6192
2	0.288675	0.816497	1.885618	2.919986	4.30265	6.96456	9.92484	31.5991
3	0.276671	0.764892	1.637744	2.353363	3.18245	4.54070	5.84091	12.9240
4	0.270722	0.740697	1.533206	2.131847	2.77645	3.74695	4.60409	8.6103
5	0.267181	0.726687	1.475884	2.015048	2.57058	3.36493	4.03214	6.8688
6	0.264835	0.717558	1.439756	1.943180	2.44691	3.14267	3.70743	5.9588
7	0.263167	0.711142	1.414924	1.894579	2.36462	2.99795	3.49948	5.4079
8	0.261921	0.706387	1.396815	1.859548	2.30600	2.89646	3.35539	5.0413
9	0.260955	0.702722	1.383029	1.833113	2.26216	2.82144	3.24984	4.7809
10	0.260185	0.699812	1.372184	1.812461	2.22814	2.76377	3.16927	4.5869
11	0.259556	0.697445	1.363430	1.795885	2.20099	2.71808	3.10581	4.4370
12	0.259033	0.695483	1.356217	1.782288	2.17881	2.68100	3.05454	4.3178
13	0.258591	0.693829	1.350171	1.770933	2.16037	2.65031	3.01228	4.2208
14	0.258213	0.692417	1.345030	1.761310	2.14479	2.62449	2.97684	4.1405
15	0.257885	0.691197	1.340606	1.753050	2.13145	2.60248	2.94671	4.0728
16	0.257599	0.690132	1.336757	1.745884	2.11991	2.58349	2.92078	4.0150
17	0.257347	0.689195	1.333379	1.739607	2.10982	2.56693	2.89823	3.9651
18	0.257123	0.688364	1.330391	1.734064	2.10092	2.55238	2.87844	3.9216
19	0.256923	0.687621	1.327728	1.729133	2.09302	2.53948	2.86093	3.8834
20	0.256743	0.686954	1.325341	1.724718	2.08596	2.52798	2.84534	3.8495
21	0.256580	0.686352	1.323188	1.720743	2.07961	2.51765	2.83136	3.8193
22	0.256432	0.685805	1.321237	1.717144	2.07387	2.50832	2.81876	3.7921
23	0.256297	0.685306	1.319460	1.713872	2.06866	2.49987	2.80734	3.7676
24	0.256173	0.684850	1.317836	1.710882	2.06390	2.49216	2.79694	3.7454
25	0.256060	0.684430	1.316345	1.708141	2.05954	2.48511	2.78744	3.7251
26	0.255955	0.684043	1.314972	1.705618	2.05553	2.47863	2.77871	3.7066
27	0.255858	0.683685	1.313703	1.703288	2.05183	2.47266	2.77068	3.6896
28	0.255768	0.683353	1.312527	1.701131	2.04841	2.46714	2.76326	3.6739
29	0.255684	0.683044	1.311434	1.699127	2.04523	2.46202	2.75639	3.6594
30	0.255605	0.682756	1.310415	1.697261	2.04227	2.45726	2.75000	3.6460
z	0.253347	0.674490	1.281552	1.644854	1.95996	2.32635	2.57583	3.2905
CI	—	—	80%	90%	95%	98%	99%	99.9%

- Example:

If you have sample of 19 patient find $P(T > 2.101)$.

Solution

$$n = 19$$

The degree of freedom = $n - 1 = 19 - 1 = 18$.

- Example:

Find $P(T > 1.50)$ with $df = 18$

Solution

From the T-table, we have

$$0.05 < P(T > 1.50) < 0.10$$

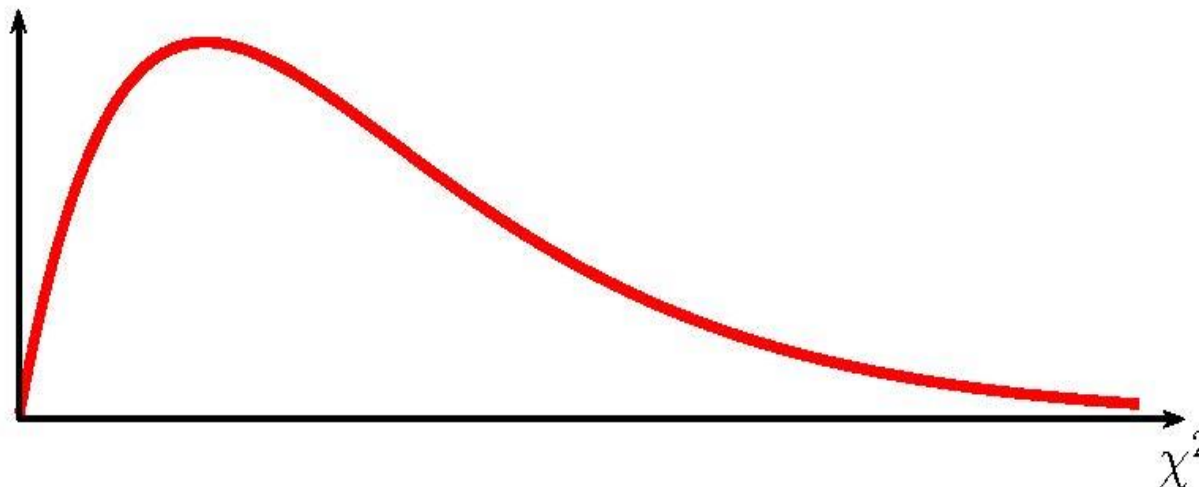
Chi-square distribution (χ^2 -distribution)

- The **Chi-square distribution** (χ^2) is a continuous probability distribution defined as the sum of the squares of independent, standard normal random variables (Z_i).

$$\chi^2 = Z_1^2 + Z_2^2 + \cdots + Z_k^2.$$

Where k is the degrees of freedom for χ^2 -distribution.

- It is always **non-negative** (0 to ∞) and **right-skewed**. As **degrees of freedom increase**, the curve becomes more symmetrical and approaches a normal distribution.
- The mean is equal to the degrees of freedom (k), and the variance is ($2k$).

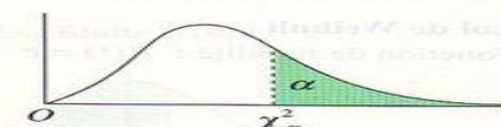


χ^2 -Distribution Table:

Table de distribution de χ^2 (loi de K. Pearson)

La table donne la probabilité α , en fonction du nombre de degrés de liberté v , pour que χ^2 égale ou dépasse une valeur donnée χ^2_α

$$\alpha = P(\chi^2 \geq \chi^2_\alpha)$$



v	$\alpha = 0,990$	$\alpha = 0,975$	$\alpha = 0,950$	$\alpha = 0,900$	$\alpha = 0,100$	$\alpha = 0,050$	$\alpha = 0,025$	$\alpha = 0,010$	$\alpha = 0,001$
1	0,0002	0,0010	0,0039	0,0158	2,71	3,84	5,02	6,63	10,83
2	0,02	0,05	0,10	0,21	4,61	5,99	7,38	9,21	13,82
3	0,12	0,22	0,35	0,58	6,25	7,81	9,35	11,34	16,27
4	0,30	0,48	0,71	1,06	7,78	9,49	11,14	13,28	18,47
5	0,55	0,83	1,15	1,61	9,24	11,07	12,83	15,09	20,52
6	0,87	1,24	1,64	2,20	10,64	12,59	14,45	16,81	22,46
7	1,24	1,69	2,17	2,83	12,02	14,07	16,01	18,47	24,32
8	1,65	2,18	2,73	3,49	13,36	15,51	17,53	20,09	26,13
9	2,09	2,70	3,33	4,17	14,68	16,92	19,02	21,67	27,88
10	2,56	3,25	3,94	4,87	15,99	18,31	20,48	23,21	29,59
11	3,05	3,82	4,57	5,58	17,27	19,67	21,92	24,72	31,26
12	3,57	4,40	5,23	6,30	18,55	21,03	23,34	26,22	32,91
13	4,11	5,01	5,89	7,04	19,81	22,36	24,74	27,69	34,53
14	4,66	5,63	6,57	7,79	21,06	23,68	26,12	29,14	36,12
15	5,23	6,26	7,26	8,55	22,31	25,00	27,49	30,58	37,70
16	5,81	6,91	7,96	9,31	23,54	26,30	28,84	32,00	39,25
17	6,41	7,56	8,67	10,08	24,77	27,59	30,19	33,41	40,79
18	7,01	8,23	9,39	10,86	25,99	28,87	31,53	34,80	42,31
19	7,63	8,91	10,12	11,65	27,20	30,14	32,85	36,19	43,82
20	8,26	9,59	10,85	12,44	28,41	31,41	34,17	37,57	45,32
21	8,90	10,28	11,59	13,24	29,61	32,67	35,48	38,93	46,80
22	9,54	10,98	12,34	14,04	30,81	33,92	36,78	40,29	48,27
23	10,20	11,69	13,09	14,85	32,01	35,17	38,08	41,64	49,73
24	10,86	12,40	13,85	15,66	33,20	36,41	39,37	42,98	51,18
25	11,52	13,12	14,61	16,47	34,38	37,65	40,65	44,31	52,62
26	12,20	13,84	15,38	17,29	35,56	38,88	41,92	45,64	54,05
27	12,88	14,57	16,15	18,11	36,74	40,11	43,19	46,96	55,48
28	13,57	15,31	16,93	18,94	37,92	41,34	44,46	48,28	56,89
29	14,26	16,05	17,71	19,77	39,09	42,56	45,72	49,59	58,30
30	14,95	16,79	18,49	20,60	40,26	43,77	46,98	50,89	59,70

Quand v est supérieur à 30, on utilise la table de la loi normale (table de l'écart réduit) avec :

$$r = \sqrt{2\chi^2} - \sqrt{2v-1}$$

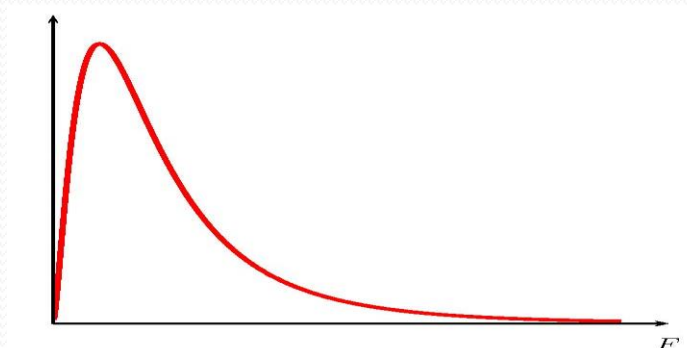
F- distribution

- The **F-distribution** (also known as the Fisher-Snedecor distribution) is a continuous probability distribution that primarily describes **the ratio of two independent chi-square random variables**, each divided by its respective degrees of freedom.
- If U_1 and U_2 are independent random variables following chi-square distributions with d_1 and d_2 degrees of freedom respectively, the random variable is defined as:

$$F = \frac{U_1/d_1}{U_2/d_2}$$

This resulting variable follows an **F-distribution** denoted as $F(d_1, d_2)$.

- Since it is a ratio of variances (or **chi-square variables**), **the F-value** is always greater than or equal to zero.
- It is always **non-negative** (0 to ∞) and **right-skewed** (positively skewed). As the degrees of freedom increase, the distribution becomes more symmetrical and eventually approximates a normal distribution.
- The mean is equal to the degrees of freedom $\left(\frac{d_2}{d_2-2}\right)$, and the variance is $\left(\frac{2d_2^2(d_1+d_2-2)}{d_1(d_2-2)^2(d_2-4)}\right)$.





Alpha = .05	1	2	3	4	5	6	7	8	9	10
1	161.4476	199.5000	215.7073	224.5832	230.1619	233.9860	236.7684	238.8827	240.5433	241.8817
2	18.5128	19.0000	19.1643	19.2468	19.2964	19.3295	19.3532	19.3710	19.3848	19.3959
3	10.1280	9.5521	9.2766	9.1172	9.0135	8.9406	8.8867	8.8452	8.8123	8.7855
4	7.7086	6.9443	6.5914	6.3882	6.2561	6.1631	6.0942	6.0410	5.9988	5.9644
5	6.6079	5.7861	5.4095	5.1922	5.0503	4.9503	4.8759	4.8183	4.7725	4.7351
6	5.9874	5.1433	4.7571	4.5337	4.3874	4.2839	4.2067	4.1468	4.0990	4.0600
7	5.5914	4.7374	4.3468	4.1203	3.9715	3.8660	3.7870	3.7257	3.6767	3.6365
8	5.3177	4.4590	4.0662	3.8379	3.6875	3.5806	3.5005	3.4381	3.3881	3.3472
9	5.1174	4.2565	3.8625	3.6331	3.4817	3.3738	3.2927	3.2296	3.1789	3.1373
10	4.9646	4.1028	3.7083	3.4780	3.3258	3.2172	3.1355	3.0717	3.0204	2.9782
11	4.8443	3.9823	3.5874	3.3567	3.2039	3.0946	3.0123	2.9480	2.8962	2.8536
12	4.7472	3.8853	3.4903	3.2592	3.1059	2.9961	2.9134	2.8486	2.7964	2.7534
13	4.6672	3.8056	3.4105	3.1791	3.0254	2.9153	2.8321	2.7669	2.7144	2.6710
14	4.6001	3.7389	3.3439	3.1122	2.9582	2.8477	2.7642	2.6987	2.6458	2.6022
15	4.5431	3.6823	3.2874	3.0556	2.9013	2.7905	2.7066	2.6408	2.5876	2.5437
16	4.4940	3.6337	3.2389	3.0069	2.8524	2.7413	2.6572	2.5911	2.5377	2.4935
17	4.4513	3.5915	3.1968	2.9647	2.8100	2.6987	2.6143	2.5480	2.4943	2.4499
18	4.4139	3.5546	3.1599	2.9277	2.7729	2.6613	2.5767	2.5102	2.4563	2.4117
19	4.3807	3.5219	3.1274	2.8951	2.7401	2.6283	2.5435	2.4768	2.4227	2.3779
20	4.3512	3.4928	3.0984	2.8661	2.7109	2.5990	2.5140	2.4471	2.3928	2.3479
21	4.3248	3.4668	3.0725	2.8401	2.6848	2.5727	2.4876	2.4205	2.3660	2.3210
22	4.3009	3.4434	3.0491	2.8167	2.6613	2.5491	2.4638	2.3965	2.3419	2.2967
23	4.2793	3.4221	3.0280	2.7955	2.6400	2.5277	2.4422	2.3748	2.3201	2.2747
24	4.2597	3.4028	3.0088	2.7763	2.6207	2.5082	2.4226	2.3551	2.3002	2.2547
25	4.2417	3.3852	2.9912	2.7587	2.6030	2.4904	2.4047	2.3371	2.2821	2.2365
26	4.2252	3.3690	2.9752	2.7426	2.5868	2.4741	2.3883	2.3205	2.2655	2.2197
27	4.2100	3.3541	2.9604	2.7278	2.5719	2.4591	2.3732	2.3053	1.7306	1.6717
28	4.1960	3.3404	2.9467	2.7141	2.5581	2.4453	2.3593	2.2913	2.2360	2.1900
29	4.1830	3.3277	2.9340	2.7014	2.5454	2.4324	2.3463	2.2783	2.2229	2.1768
30	4.1709	3.3158	2.9223	2.6896	2.5336	2.4205	2.3343	2.2662	2.2107	2.1646
40	4.0847	3.2317	2.8387	2.6060	2.4495	2.3359	2.2490	2.1802	2.1240	2.0772
60	4.0012	3.1504	2.7581	2.5252	2.3683	2.2541	2.1665	2.0970	2.0401	1.9926
120	3.9201	3.0718	2.6802	2.4472	2.2899	2.1750	2.0868	2.0164	1.9588	1.9105
INF	3.8415	2.9957	2.6049	2.3719	2.2141	2.0986	2.0096	1.9384	1.8799	1.8307

Sampling distribution

- ❑ **Population:** It is a large group of people or element which we are interest.
- ❑ **Sample:** It is a subset of population.
- ❑ **Statistics:** Is a random variable calculated from the sample.
- ❑ **Parameter:** Is a random variable calculated from the population.
- ❑ **Sampling distribution:** Is the probability distribution of any statistics.
- ❑ **Remark:**

Population symbols	Sample symbols
$N \rightarrow$ population size	$n \rightarrow$ sample size
$\mu \rightarrow$ population mean	$\bar{x} \rightarrow$ sample mean
$\sigma^2 \rightarrow$ population variance	$s^2 \rightarrow$ sample variance
$\sigma \rightarrow$ population standard deviation	$s \rightarrow$ sample standard deviation

Sampling distribution of the mean ($\bar{X}$)

- Let $X_1, X_2, \dots, X_n$ is a random sample of size n taken from population with mean μ and variance σ^2 then:

$$E(\bar{X}) = \mu_{\bar{x}} = \mu$$

and

$$V(\bar{X}) = \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$$

- **Remark:** We note that the quantity:

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

is called the standard error and we note that :

- ✓ As sample size increases, the standard error decrease and thus the variability of sample means decreases.

Theorem:

Let $X_1, X_2, \dots, X_n$ is a random sample of size n taken from normal population with mean μ and variance σ^2 then the sampling distribution of $\bar{X}$ will be normal with mean $\mu_{\bar{x}} = \mu$ and variance $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$ i.e.

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

Central limit theorem:

Let $X_1, X_2, \dots, X_n$ is a random sample of large enough size n taken from population with mean μ and variance σ^2 then the sampling distribution of $\bar{X}$ will be approximately normal with mean $\mu_{\bar{x}} = \mu$ and variance $\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$ i.e.

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

Very Important Remark:

- Let $X_1, X_2, \dots, X_n$ is a random sample of size n taken from population with mean μ and variance σ^2 (Known), then

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

If the population is normally distributed or the sample size $n \geq 30$.

- Let $X_1, X_2, \dots, X_n$ is a random sample of size n taken from population with mean μ and variance σ^2 (Unknown), then

$$Z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim N(0, 1)$$

If the sample size $n \geq 30$.

- Let $X_1, X_2, \dots, X_n$ is a random sample of size n taken from population with mean μ and variance σ^2 (Unknown), then

$$T = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim T_{(n-1)}$$

If the population is approximately normal distributed and the sample size $n < 30$.

- Example:

A sample of size **16** is drawn from a normal population with mean **200** and variance **36**. Find the probability that the sample mean will be less than **199**.

Solution

$$n = 16, \quad \mu = 200, \quad \sigma^2 = 36, \quad \sigma = 6$$

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

$$P(\bar{X} \leq 199) = P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{199 - 200}{\frac{6}{\sqrt{16}}}\right) = P(Z \leq -0.67)$$

$$= \varphi(-0.67) = 1 - \varphi(0.67) = 1 - 0.7486 = 0.251$$

- Example:

An electric firm manufactures light bulb that have a length of life that approximately normally distributed with mean equal to **800** hours and standard deviation of **40** hours. Find the probability that a random sample of **16** bulbs will have an average life of less than **775** hours.

Solution

$$\mu = 800,$$

$$\sigma = 40,$$

$$n = 16$$

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

$$P(\bar{X} \leq 775) = P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{775 - 800}{\frac{40}{\sqrt{16}}}\right) = P(Z \leq -2.5)$$

$$= \varphi(-2.5) = 1 - \varphi(2.5) = 1 - 0.9938 = 0.006$$

- Example:

A sample of size **64** is drawn from a population with mean **3.2** and standard deviation **1.6**. find the probability that the sample mean will be :

- a) More than **3.5** .
- b) Less than **2.7** .

Solution

$$n = 64, \quad \mu = 3.2, \quad \sigma = 1.6$$

$$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \sim N(0, 1)$$

$$a) P(\bar{X} > 3.5) = P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} > \frac{3.5 - 3.20}{\frac{1.6}{\sqrt{64}}}\right) = P(Z > 1.5) = 1 - \varphi(1.5) = 1 - 0.9332 = 0.0668$$

$$b) P(\bar{X} < 2.7) = P\left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} < \frac{2.7 - 3.20}{\frac{1.6}{\sqrt{64}}}\right) = P(Z < -2.5) = \varphi(-2.5) = 1 - \varphi(2.5) = 1 - 0.9938 = 0.0062$$

- Example:

A sample of size **64** has standard deviation **5** drawn from population with mean **3.2**. Find the probability that the sample mean will be:

- a) More than **2.2** .
- b) Less than **1.7** .

Solution

$$n = 64, \quad \mu = 3.2, \quad s = 5$$

$$Z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} \sim N(0, 1)$$

$$\begin{aligned} a) P(\bar{X} > 2.2) &= P\left(\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} > \frac{2.2 - 3.2}{\frac{5}{\sqrt{64}}}\right) = P(Z > -1.6) = 1 - \varphi(-1.6) = 1 - (1 - \varphi(1.6)) = \\ &= 1 - (1 - 0.9452) = 0.9452. \end{aligned}$$

$$b) P(\bar{X} < 1.7) = P\left(\frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} < \frac{1.7 - 3.2}{\frac{5}{\sqrt{64}}}\right) = P(Z < -2.4) = \varphi(-2.4) = 1 - \varphi(2.4) = 1 - 0.9918 = 0.0082$$

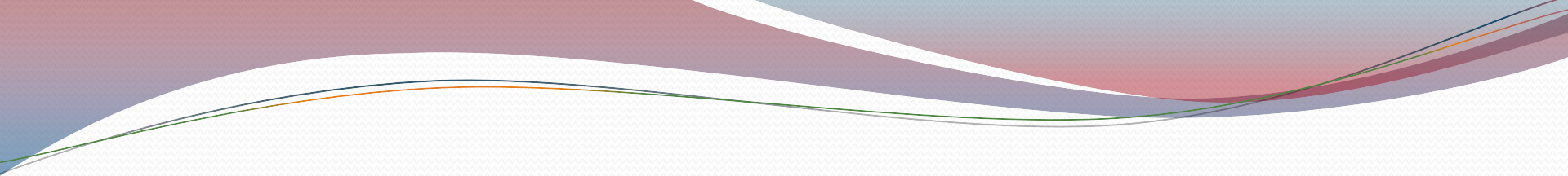
- Example:

Certain Tubes manufactured by a company have a mean lifetime of **800** hours and standard deviation of **40** hours. Find the probability that a random sample of **16** tubes taken from the group will have mean lifetime :

- a) More than **815** hours .
- b) Between **790** and **810** hours .

Solution

حل بنفسك.....



Thank you for your
Attention!

